

Herramientas de Gestión y Auditoría

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Administración de Proyectos de Software
Semestre 2026-1

1 Taiga

2 Sonar

3 Importancia en la Ingeniería de Software

¿Qué es Taiga?

- Plataforma de **gestión ágil de proyectos**.
- Apoya metodologías como **Scrum y Kanban**.
- Permite organizar tareas, backlog y sprints.

Características principales de Taiga

- Interfaz visual con tableros y paneles.
- Open Source, flexible y colaborativa.
- Gestión de backlog, épicas e historias de usuario.
- Soporte de múltiples roles de equipo.

- Crear proyectos con tableros de tareas.
- Definir historias de usuario y sprints.
- Asignar responsables y monitorear el avance.
- Integrar con repositorios (GitHub, GitLab).



TAIGA

Features Pricing About Community

Log in

Get Taiga

Taiga: *The* free and open-source project management tool

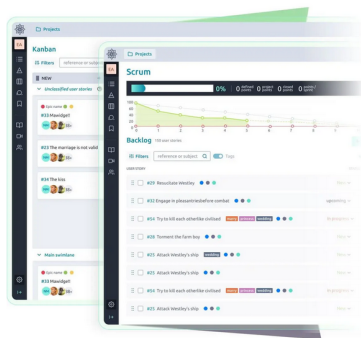
For cross-functional agile teams to work effectively

A featured-rich software that offers a very simple start through its intuitive user interface.

Get Taiga

Self hosted

tree.taiga.io/login *agile with Taiga already. Join them!*



Ejemplo

 Projects

 Corazón contenido

 Scrum

 Backlog

 Issues

 Search

 Wiki

 Team

 Settings

collapse menu

Scrum

0% 74 defined points 0 closed points 0 points / sprint



CUSTOMIZE YOUR BACKLOG GRAPH

To have a nice graph that helps you follow the evolution of the project you have to set up the points and sprints through the [Admin](#)

Backlog

10 user stories

+ USER STORY

Filters

subject or reference

Tags

USER STORY	STATUS	POINTS
☐ #1 HP-001	New	8
☐ #2 HP-002	New	11
☐ #3 HP-003	New	4
☐ #4 HP-004	New	14
☐ #5 HP-005	New	13

SPRINTS



There are no sprints yet
[Add a sprint +](#)



Ejemplo

The screenshot shows a Jira project page for 'Corazón contento'. The left sidebar contains navigation links: 'Corazón contento', 'Scrum', 'Issues', 'Search', 'Wiki', 'Team', 'Settings', and a 'collapse menu' button. The main content area features the project name 'Corazón contento' with a pixelated heart icon. Below the name is a description: 'Plan de Desarrollo de Software para una aplicación que busca reducir el desperdicio alimentario, conectando a negocios con consumidores para reubicar alimentos sobrantes. Los objetivos incluyen alcanzar 10,000 usuarios activos y reducir el desperdicio en un 20% en el primer año.' To the right of the description are interaction buttons: 'Like' (0), 'Watching' (1), and an email icon. A list of updates follows, each starting with a user icon and the text 'Fer Ordoñez has updated'. The updates include changes to 'Design' and 'UX' points for various user stories (e.g., #8 HC-003, #7 HC-002, #7 HC-002, #6 HC-001, #5 HP-005, #4 HP-004, #4 HP-004, #4 HP-004, #4 HP-004). Each update is timestamped as '7 days ago' or '8 days ago'. On the right side of the updates list is a 'Team' section with a team icon.

Projects

Corazón contento

Scrum

Issues

Search

Wiki

Team

Settings

collapse menu

Corazón contento

Plan de Desarrollo de Software para una aplicación que busca reducir el desperdicio alimentario, conectando a negocios con consumidores para reubicar alimentos sobrantes. Los objetivos incluyen alcanzar 10,000 usuarios activos y reducir el desperdicio en un 20% en el primer año.

Like 0

Watching 1

Fer Ordoñez has updated 'Design' points of the user story #8 HC-003 to 10 7 days ago

Fer Ordoñez has updated 'UX' points of the user story #7 HC-002 to 10 7 days ago

Fer Ordoñez has updated 'Design' points of the user story #7 HC-002 to 2 7 days ago

Fer Ordoñez has updated 'Design' points of the user story #6 HC-001 to 2 7 days ago

Fer Ordoñez has updated 'UX' points of the user story #5 HP-005 to 13 7 days ago

Fer Ordoñez has updated 'UX' points of the user story #4 HP-004 to 10 7 days ago

Fer Ordoñez has updated 'Back' points of the user story #4 HP-004 to 2 8 days ago

Fer Ordoñez has updated 'Front' points of the user story #4 HP-004 to 2 8 days ago

Fer Ordoñez has updated 'UX' points of the user story #4 HP-004 to 3 8 days ago

Team

Ejemplo

#1 HP-001

USER STORY

Add tag +

Created by Fer Ordoñez
03 Oct 2025 17:39

Como Productor de comida, quiero publicar alimentos sobrantes con Foto, descripción, fecha y precio, para no perderlos antes de que caduquen

0 Attachments

+

OPEN NEW ~

POINTS

UX	3
Design	?
Front	3
Back	2
total points	8

Beneficios de Taiga

- Gratuita y personalizable frente a Jira o Asana.
- Fácil de usar y con curva de aprendizaje baja.
- Pensada para equipos pequeños y medianos.
- Permite control de proyectos sin costos altos.

¿Qué es Sonar?

- Herramienta de **auditoría y análisis estático de código**.
- Evalúa calidad, seguridad y mantenibilidad.
- Disponible como SonarQube (local) o SonarCloud (SaaS).

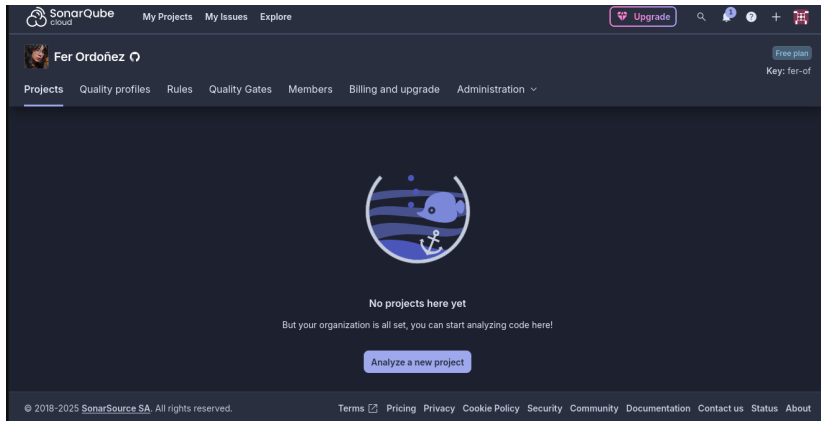
Características principales de Sonar

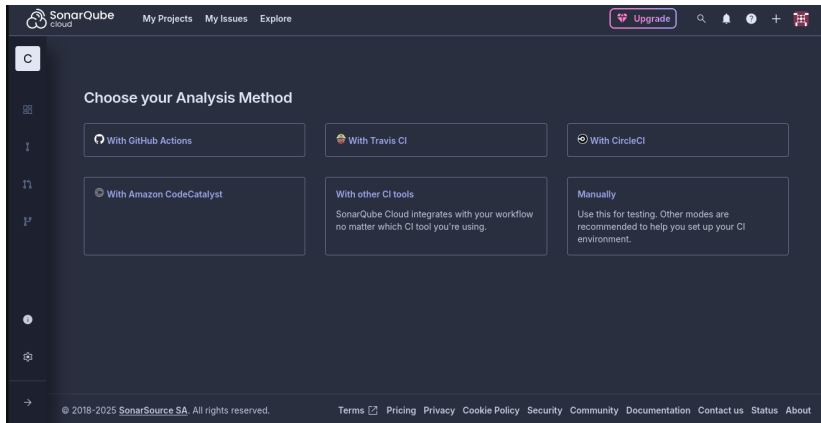
- Soporta múltiples lenguajes (Java, Python, C#, JS, etc.).
- Ofrece **Quality Gates** con métricas claras.
- Detecta bugs, vulnerabilidades y malas prácticas.
- Integración con CI/CD y repositorios.

Que lograremos?

- Una cuenta en SonarCloud (donde verás la calidad de tu código).
- Un token de autenticación seguro.
- Un archivo de configuración en tu proyecto (sonar-project.properties).
- Un workflow de GitHub Actions que ejecutará el análisis automático cada vez que hagas un push o pull request.

Ejemplo





The screenshot shows the SonarQube cloud interface. The top navigation bar includes the SonarQube logo, 'My Projects', 'My Issues', and 'Explore'. A red 'Upgrade' button is visible in the top right. The breadcrumb trail reads: 'Fer Ordoñez > corazonContento > Administration / Analysis Method > Analyze a project with GitHub Actions'.

Analyze a project with a GitHub Action

1 Create a GitHub Secret

In your GitHub repository, go to [Settings > Secrets and variables > Actions](#) and create a new secret with the following details:

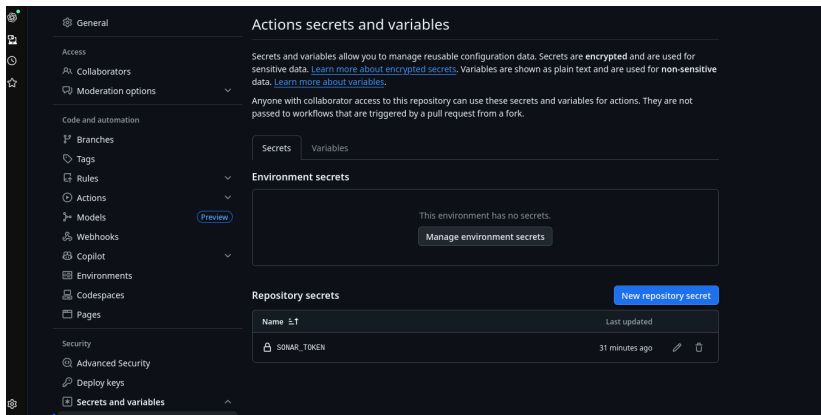
- 1 In the Name field, enter
- 2 In the Value field, enter

2 Create or update a build file

What option best describes your project?

Which OS do you run your build on?

Ejemplo



The screenshot shows the GitHub Actions 'Secrets and variables' page. The left sidebar contains a navigation menu with categories: General, Access, Code and automation, and Security. The 'Secrets and variables' option is selected under the Security category. The main content area is titled 'Actions secrets and variables' and includes a description of secrets and variables, tabs for 'Secrets' and 'Variables', and sections for 'Environment secrets' (showing no secrets) and 'Repository secrets' (showing a secret named SONAR_TOKEN).

General

Access

- Collaborators
- Moderation options

Code and automation

- Branches
- Tags
- Rules
- Actions
- Models Preview
- Webhooks
- Copilot
- Environments
- Codespaces
- Pages

Security

- Advanced Security
- Deploy keys
- Secrets and variables**

Actions secrets and variables

Secrets and variables allow you to manage reusable configuration data. Secrets are **encrypted** and are used for sensitive data. [Learn more about encrypted secrets](#). Variables are shown as plain text and are used for non-sensitive data. [Learn more about variables](#).

Anyone with collaborator access to this repository can use these secrets and variables for actions. They are not passed to workflows that are triggered by a pull request from a fork.

Secrets **Variables**

Environment secrets

This environment has no secrets.

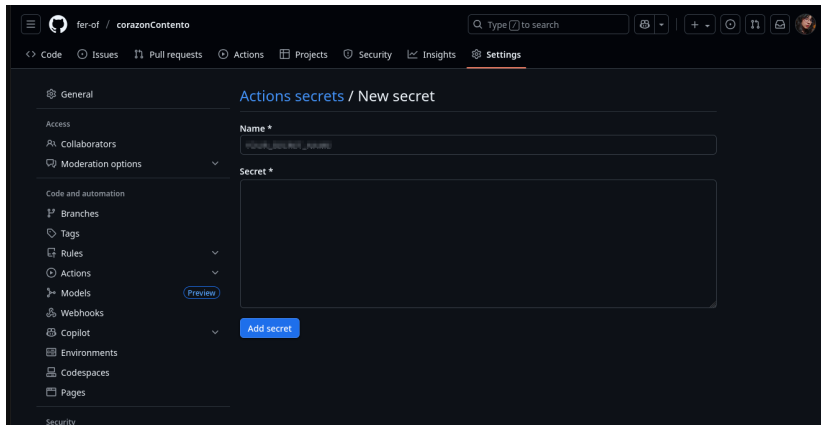
[Manage environment secrets](#)

Repository secrets

[New repository secret](#)

Name	Last updated
SONAR_TOKEN	31 minutes ago

Ejemplo



Ejemplo

3 Create a `sonar-project.properties` file

Create a configuration file in the root directory of the project and name it `sonar-project.properties`

```
sonar.projectKey=fer-of_corazonContento  
sonar.organization=fer-of
```

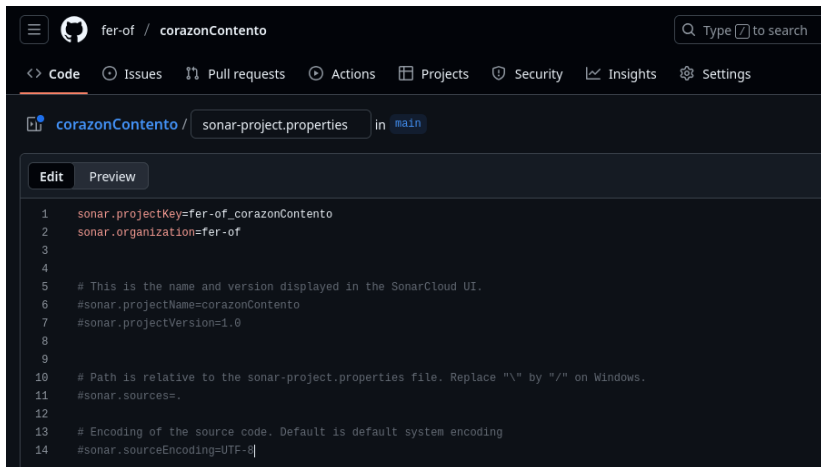
Copy

```
# This is the name and version displayed in the SonarCloud UI.  
#sonar.projectName=corazonContento  
#sonar.projectVersion=1.0
```

```
# Path is relative to the sonar-project.properties file. Replace "\" by "/" on Windows.  
#sonar.sources=.
```

```
# Encoding of the source code. Default is default system encoding  
#sonar.sourceEncoding=UTF-8
```

Ejemplo



The screenshot shows a GitHub repository for 'fer-of / corazonContento'. The 'Code' tab is selected, displaying the 'sonar-project.properties' file in the 'main' branch. The file content is as follows:

```
1 sonar.projectKey=fer-of_corazonContento
2 sonar.organization=fer-of
3
4
5 # This is the name and version displayed in the SonarCloud UI.
6 #sonar.projectName=corazonContento
7 #sonar.projectVersion=1.0
8
9
10 # Path is relative to the sonar-project.properties file. Replace "\" by "/" on Windows.
11 #sonar.sources=.
12
13 # Encoding of the source code. Default is default system encoding
14 #sonar.sourceEncoding=UTF-8
```

Ejemplo

Create or update your `.github/workflows/build.yml` 

Here is a base configuration to run a SonarQube Cloud analysis on your master branch and Pull Requests. If you already have some GitHub Actions, you might want to just add some of these new steps to an existing one.

```
name: Build
on:
  push:
    branches:
      - main
  pull_request:
    types: [opened, synchronize, reopened]
jobs:
  sonarqube:
    name: SonarQube
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
        with:
          fetch-depth: 0 # Shallow clones should be disabled for a better relevancy of analysis
      - name: SonarQube Scan
        uses: SonarSource/sonarqube-scan-action@v6
        env:
          SONAR_TOKEN: ${ secrets.SONAR_TOKEN }
```

 Copy

Ejemplo

fer-of / corazonContento

Code Issues Pull requests Actions Projects Security Insights Settings

corazonContento / .github / workflows / build.yml in main

Cancel changes Commit changes...

Edit Preview Spaces 2 No wrap

```
1 name: Build
2 on:
3   push:
4     branches:
5       - main
6   pull_request:
7     types: [opened, synchronize, reopened]
8 jobs:
9   sonarqube:
10    name: SonarQube
11    runs-on: ubuntu-latest
12    steps:
13      - uses: actions/checkout@v4
14        with:
15          fetch-depth: 0 # Shallow clones should be disabled for a better relevancy of analysis
16      - name: SonarQube Scan
17        uses: SonarSource/sonarqube-scan-action@v6
18        env:
19          SONAR_TOKEN: ${{ secrets.SONAR_TOKEN }}
```

Use **Control + Shift + Tab** to toggle the **tab** key moving focus. Alternatively, use **esc** then **tab** to move to the next interactive element on the page.

Use **Control + space** to trigger autocomplete in most situations.

Marketplace Documentation

Search Marketplace for Actions

Featured Actions

- Setup Node.js environment** 4.4k
By actions
Setup a Node.js environment by adding problem matchers and optionally downloading and adding it to the PATH
- Setup Java JDK** 1.6k
By actions
Set up a specific version of the Java JDK and add the command-line tools to the PATH
- Download a Build Artifact** 1.7k
By actions
Download a build artifact that was previously uploaded in the workflow by the upload-artifact action
- Close Stale Issues** 1.6k
By actions

Ejemplo

SonarQube cloud

My Projects My Issues Explore

Upgrade

Fer Ordoñez > corazonContento > main

Summary Issues Security Hotspots Measures Code Activity

Main Branch Summary

19 Lines of Code • Last analysis 1 minute ago • 8ca1f7de

Quality Gate: [Sonar way](#)

Not computed

Next scan will generate a Quality Gate.

Security	Reliability	Maintainability
0 Open Issues A	0 Open Issues A	0 Open Issues A
Accepted Issues	Coverage	Duplications

Ejemplo

The screenshot displays the SonarQube cloud web interface. At the top, the navigation bar includes 'My Projects', 'My Issues', and 'Explore'. A pink 'Upgrade' button is visible in the top right corner. The main content area shows the 'Main Branch Summary' for a project named 'Fer Ordoñez > corazonContento'. The summary indicates '19 Lines of Code' and 'Last analysis 27 seconds ago'. A large green checkmark and the word 'Passed' confirm the quality gate status. Below this, a table provides a detailed breakdown of the quality metrics:

Category	Value	Grade
Security	0 Open Issues	A
Reliability	0 Open Issues	A
Maintainability	0 Open Issues	A

- Integrar análisis en cada commit o pull request.
- Configurar métricas de calidad mínimas.
- Generar reportes automáticos para el equipo.
- Notificar resultados en Slack u otros canales.

Beneficios de Sonar

- Open Source en su versión básica.
- Comparado con ESLint o Checkmarx: ofrece mayor cobertura.
- Mejora la calidad de software desde etapas tempranas.
- Refuerza la cultura DevOps con auditoría continua.

Importancia de Taiga y Sonar

- **Taiga:** asegura la correcta gestión ágil del proyecto.
- **Sonar:** garantiza calidad y seguridad del código.
- Juntas fortalecen la **ingeniería de software**:
 - Organización eficiente del trabajo.
 - Prevención de errores y reducción de riesgos.
 - Soporte al ciclo completo de desarrollo.

- Taiga y Sonar son herramientas clave en la administración moderna de proyectos de software.
- Su combinación ejemplifica la unión de **gestión ágil** y **auditoría técnica**.
- Aportan valor tanto en entornos académicos como profesionales.